
SYSTEM VULNERABILITY CHECKLIST

A Proactive Hardening & Risk Assessment Report

DecodeLabs Cyber Security Internship

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1. Introduction

A vulnerability is any weakness in a system's configuration that an attacker - or simple bad luck - can exploit. Proactive hardening means finding and fixing these gaps before an incident occurs, instead of reacting after the damage is done. This report documents a defensive security audit performed on a personal Windows laptop, covering identity controls, patch status, disk encryption, network protection, and user-practice hygiene.

Each check was performed with built-in Windows tools only, following the principle that a useful audit must be repeatable by a non-expert. Every check ends in a definitive PASS or FAIL, the failures are ranked by risk, and a concrete remediation plan is provided.

2. Audit Methodology

The audit combined command-line verification with manual settings review:

1. **Network protection:** the state of all three Windows Firewall profiles was verified in an elevated PowerShell session using Get-NetFirewallProfile.
2. **Disk encryption:** volume encryption status was queried using Get-BitLockerVolume.
3. **Patch status:** update state was reviewed under Settings > Windows Update.
4. **Identity and user practices:** sign-in options, screen-lock timeout, and multi-factor authentication on key online accounts were reviewed manually.

Screenshots of the PowerShell outputs are included in the repository as evidence of the audit.

3. Checklist Results

Sr	Security Check	Audit Method	Observed Result	Status
1	Host firewall enabled on all profiles	PowerShell: Get-NetFirewallProfile	Domain, Private and Public profiles all Enabled = True	PASS
2	Full-disk encryption (BitLocker)	PowerShell: Get-BitLockerVolume	C: and D: volumes FullyDecrypted, Protection Status: Off	FAIL
3	Operating system receiving security updates	Settings > Windows Update	Windows 10 has reached end of support; device is no longer receiving security updates	FAIL
4	Login authentication	Settings > Accounts >	No PIN or password	FAIL

	(password / PIN)	Sign-in options	configured for sign-in	
5	Automatic screen lock	Settings > Power & sleep	Screen timeout set to 5 minutes	PASS
6	Multi-factor authentication on key accounts	Manual review of email / GitHub sign-in	MFA enabled - verification code required at login	PASS

Summary: 3 of 6 checks passed. The failures are analyzed and ranked below.

4. Risk-Ranked Findings

Risk Level	Finding	Potential Impact
HIGH	Operating system end of support (no security updates)	Every newly discovered Windows vulnerability remains permanently unpatched, leaving the system exposed to malware, ransomware, and remote exploits.
HIGH	No login password / PIN	Anyone with physical access to the laptop gains immediate access to all files, browsers, and saved sessions - no attack skill required.
MEDIUM	BitLocker disk encryption disabled	If the laptop is lost or stolen, the hard drive can be removed and read directly, exposing all stored data even without the Windows login.
LOW	-	Firewall, screen lock, and MFA controls are correctly configured and reduce the overall attack surface.

5. Remediation Plan

The following fixes are listed in priority order:

1. **Restore security updates (HIGH - immediate).** Enroll the device in Extended Security Updates from the Windows Update page, or upgrade to a supported operating system version. An unpatched OS undermines every other control on this list.
2. **Set a login password and PIN (HIGH - immediate).** Settings > Accounts > Sign-in options. Use a long passphrase rather than a short complex password, and add a PIN for convenient daily sign-in. This closes the physical-access gap at zero cost.
3. **Enable disk encryption (MEDIUM - this week).** Turn on BitLocker (or Device Encryption on Windows Home) once sign-in authentication is configured, and store the recovery key safely offline - never in a plain file on the same machine.
4. **Maintain the passing controls (ONGOING).** Keep all firewall profiles enabled, keep the 5-minute screen lock, and keep MFA active on email and GitHub accounts. Review this checklist monthly.

6. Conclusion

This audit demonstrates the core lesson of proactive hardening: serious vulnerabilities are usually not exotic exploits but ordinary configuration gaps - a missing password, an unpatched system, an unencrypted disk. Each finding here is fixable in minutes, yet each one, left open, offers an attacker a complete path to the system's data.

By documenting the checks, ranking the risks, and defining the fixes, this project converts theoretical security awareness into a repeatable defensive routine. Combined with the previous projects - password validation, encryption fundamentals, and phishing triage - it completes the defensive foundation of a security analyst: identify the gap, measure the risk, close it, and verify again.